

Chaiyong Mun

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EDUCATION

Seoul National University

Integrated M.S./Ph.D. program in the Department of Intelligence and Information.

– GPA: 4.0/4.3

Seoul, South Korea

2023 Fall–Present

Seoul National University

B.S. in Electrical and Computer Engineering.

– GPA: 3.5/4.3

Seoul, South Korea

2017–2023 Fall

RESEARCH INTERESTS

- Robot Manipulation
- Multimodal Imitation Learning
- Vision-Language-Action Models

EXPERIENCE

Seoul National University/DYROS Lab

Research Assistant/Dynamic Robotic Systems Laboratory

Seoul, South Korea

2023–Present

- Conducted research on robot manipulation, visual–tactile perception, robot learning, and humanoid teleoperation.
- Participated in multiple industry-sponsored research projects with Samsung Electronics and the Ministry of Trade, Industry and Energy.

PUBLICATIONS

- [1] **C. Mun**, E. Sung, H. Lee, and J. Park, “Hierarchical reactive grasping for contact-rich manipulation via topological visuo-tactile learning”, *Accepted for Publication at IEEE International Conference on Intelligent Robots and Systems (IROS) 2026*, 2026.
- [2] M. Suh, **C. Mun**, H. Lee, and J. Park, “Grasp state estimation via 3d mean resultant length”, *Proceedings of the Korea Robotics Society Annual Conference (KRoC)*, 2026.
- [3] E. Sung, S. You, J. Shin, S. Park, S. Hur, J. Kim, S. Byun, **C. Mun**, H. Lee, D. Kim, and J. Park, “Snu-avatar haptic arm: A hybrid qdd/dd operator interface for teleoperation”, *IEEE Access*, vol. 14, pp. 57 409–57 426, 2026.

PROJECTS

Multimodal Perception for Deformable Object Manipulation(Samsung DA, Project Lead, 2026)

Developed an autonomous laundry handling system for domestic service robots by integrating visual perception, tactile sensing, and closed-loop failure recovery. The system employs domain-conditioned segmentation and one-class tactile anomaly detection to enable reliable manipulation of highly deformable garments.

Learning-based Humanoid Robot Manipulation (MOTIE, Research Assistant, 2024–2025)

Developed a learning-based robot manipulation framework that integrates visuo-tactile imitation learning to generate end-effector pose and force commands for robust manipulation.

Haptic Teleoperation for Humanoid Robot Simulation (DYROS, Project Lead, 2024)

Developed a MuJoCo simulation environment for a laboratory humanoid robot (TOCABI) and implemented haptic teleoperation for object grasping and manipulation.

Transparent Object Grasp Pose Estimation (Samsung SAIT, Research Assistant, 2023)

Developed a vision-based transparent object grasp pose estimation framework by integrating RGB-D depth restoration and NeRF-based perception for reliable robotic manipulation.

RESEARCH OUTPUTS

Journal Manuscript in Preparation (First Author)

Visual–Tactile Perception and Manipulation for Deformable Objects.

Patent Application in Preparation

Dual-channel tactile anomaly detection for grasp state estimation.

TEACHING

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|---|------|
| • Teaching Assistant at Seoul National University | 2025 |
| <i>DYROS Robotics Boot Camp II (Robot Design, SI, Simulation and Control)</i> | |
| • Teaching Assistant at Seoul National University | 2025 |
| <i>DYROS Robotics Boot Camp I (ROS: Robot Operating System)</i> | |

SKILLS

- **Software:** C++, Python, MATLAB, PyTorch, Anaconda, Docker, C.
- **System Integration:** RTLinux, ROS1, ROS2, EtherCAT.
- **Applications:** Isaac Gym/Lab, MuJoCo, CoppeliaSim, MoveIt, Git, VS code, Notion.

LANGUAGES

- **Language:** Korean (mother tongue)
- **Language:** English (intermediate)